SALEEM MOHAMMED ALI

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SKILLS

CAD: SOLIDWORKS, Finite Element Analysis (FEA), Abaqus, AutoCAD, Autodesk Revit, GD&T, SketchUp
Manufacturing: DFMA, Rapid Prototyping, 3D Printing, Welding, Laser Cutting, Power Tools, Machine Shop Tools
Software / Programming: C++, Arduino, Python, PLC, MATLAB, Excel, Word, RobotC, JavaScript, HTML

EXPERIENCE

Structural Designer & Fabricator May 2025 – Present
UW WATurbine Waterloo, ON

- Designing and fabricating structural and mechanical components for a small wind turbine to participate in the International Small Wind Turbine Content (ISWTC) annually held in the Netherlands, tailoring the turbine's design for sustainability.
- Completed the fabrication of the main structural components of the turbine including the base, tower, nacelle and tailfin using a variety of machine shop tools; now researching wind turbine literature to find solutions for arising problems.
- Aiding in supervision of the testing and data collection phase, to make improvements on the turbine where necessary.

Materials Engineering Intern September 2024 – December 2024
Dana Incorporated Power Technologies R&D Centre Oakville, ON

- Conducted 2000+ mechanical tests including tensile, lap shear, peel, hardness, and corrosion, using a wide variety of machines and alloys of both aluminum and copper to analyze the mechanical properties of the different alloy compositions.
- Experimented with the reliability of different brazing techniques on aluminum with both raw material and small battery coolers, applying the DOE methods to conclude the optimal joining technique to be used in large-scale manufacturing.
- Solved manufacturing issues when they arose using characterization and quality methods, minimizing production losses.
- Maintained production schedule by diagnosing points of failure of parts in Dana manufacturing plants globally.

Field Project Coordinator January 2024 – April 2024
EllisDon – St. Michaels Hospital Project Toronto, ON

- Directed/assisted site coordination, overseeing the completion of excavation and start of foundation rebar and concrete.
- Facilitated concrete pours including foundation raft pours totaling 1100+ m^3 over 10+ hours minimizing waste expenditure.
- Assisted the senior surveyor in the process of laying out sites creating as-builts ensuring the shop drawings were followed.
- Produced and managed field-tracking tools in Excel, tracking important data integral in keeping the field running.
- Immensely refined transferable skills such as leadership, responsibility, communication and technological skills.

Vehicle Chassis Technician September 2022 – January 2023
UW Formula Motorsports Team Waterloo, ON

- Assembled an anterior crash structure (IA) supporting vehicle safety using metal bonding techniques.
- Fabricated 30+ vehicle components and constructed a jig for the chassis over to prepare the vehicle for the welding process.
- Developed essential skills in design ideation and problem-solving to work collaboratively in an automotive environment.

ADDITIONAL EXPERIENCE

Velocity Health Challenge February 2025
Velocity Incubator Waterloo, ON

- Developed an innovative safer mobility solution collaborating in a team of five for bariatric patients at Grand River Hospital.

CAD Designathon January 2024
McMaster Design League Hamilton, ON

- Designed and presented an advanced insulin cartridge holder with user-focused design in a 36-hour team competition.

PROJECTS

Guitar Robot

- Collaborated with a team of four to design and program a LEGO Mindstorms EV3 robot in C++ to play simplified guitar songs, enhancing functionality with 3D-printed and LEGO components through decision matrices and sketches to guide decisions.

Pneumatic Can Crushing Toy

- Designed & manufactured a pneumatic-hydraulic can crusher and zipline in a team of four using AutoCAD and SOLIDWORKS; delegated tasks and applied functional analysis throughout iterative prototyping to meet project objectives over 10 weeks.

EDUCATION

University of Waterloo – Candidate B.A.Sc. in Mechanical Engineering, Honours Co-op **September 2022 – April 2027**